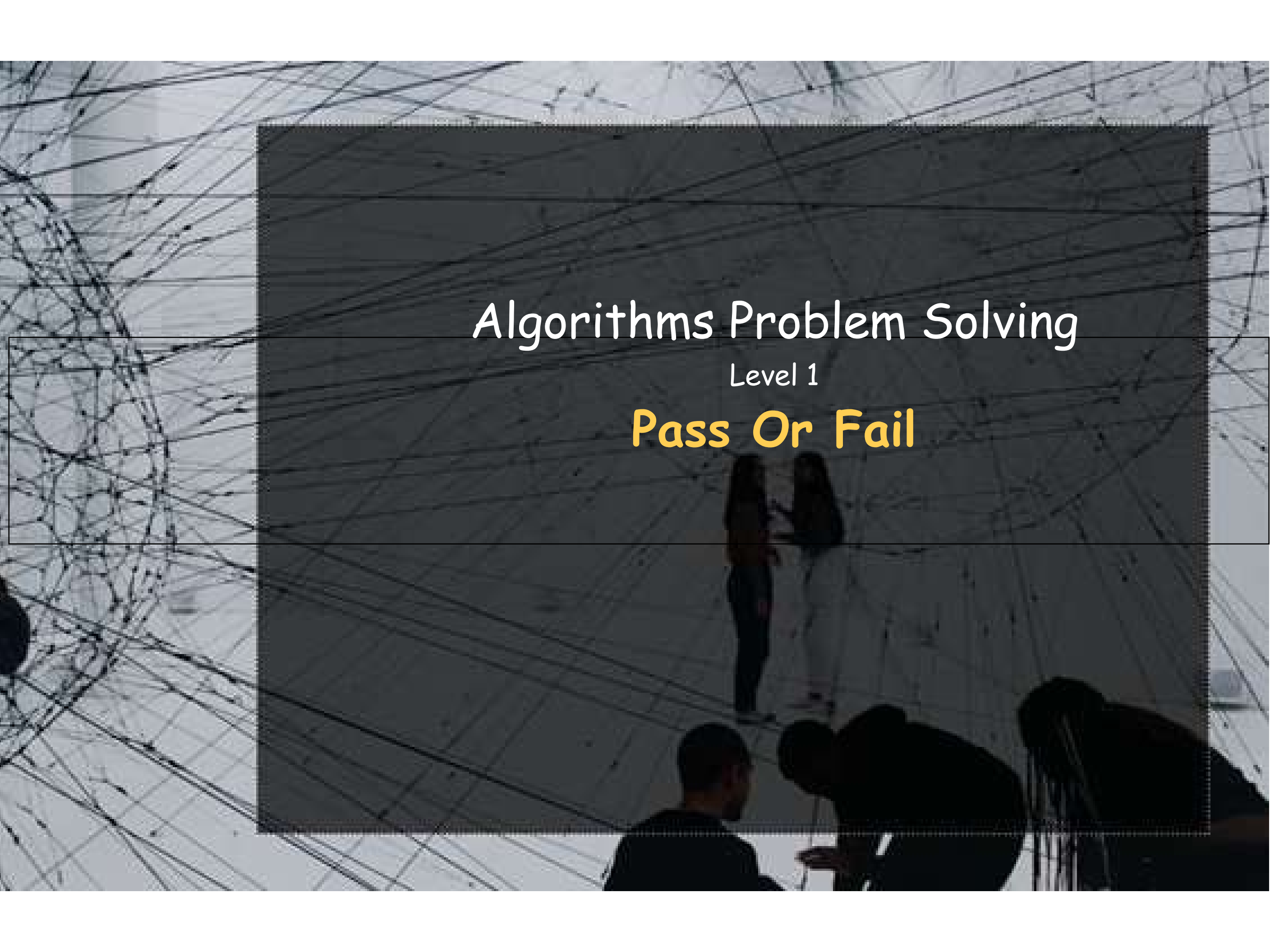


Algorithms Problem Solving

Level 1

Pass Or Fail

The background of the slide is a photograph of a basketball court. A large, dark, semi-transparent rectangle is centered over the image. The text is placed within this rectangle. In the lower part of the image, outside the dark rectangle, the silhouettes of several people are visible, some standing and some sitting, suggesting a group activity or a class setting.

Algorithms Problem Solving

Level 1

Pass Or Fail

Definition

- ◆ This program reads a student's mark and checks whether the student has passed or failed.
- ◆ The student passes if the mark is 50 or more.
- ◆ The program uses an enum, functions, and an if-else statement to determine and print the result.

Explanation

◆ The enum `enPassFail` defines two possible results:

`Pass = 1`

`Fail = 0`

◆ `readMark()` reads the student's mark and returns it.

◆ `checkPassFail()` checks whether the mark is greater than or equal to 50.

◆ If `mark >= 50`, the function returns `Pass`.

◆ Otherwise, it returns `Fail`.

◆ `printResult()` receives the mark and calls `checkPassFail()` to determine the result.

◆ If the returned value is `Pass`, it prints "You have passed."

◆ Otherwise, it prints "You have failed."

◆ In `main()`, the functions are combined:

`printResult(readMark());`

Steps in main()

1. Read the student's mark.
2. Return the mark from readMark().
3. Pass the mark to printResult().
4. Call checkPassFail() to check the mark.
5. If the mark is 50 or more, return Pass.
6. Otherwise, return Fail.
7. Print the appropriate result.

💡 Example

◆ If the user enters:

75

◆ Since $75 \geq 50$, `checkPassFail()` returns Pass.

➡ Result

You have passed.

◆ If the user enters:

45

◆ Since $45 < 50$, `checkPassFail()` returns Fail.

➡ Result

You have failed.

Note

- ◆ The condition `mark >= 50` means that exactly 50 is considered a passing mark.
- ◆ The enum provides named values, `Pass` and `Fail`, instead of using plain numbers directly.

⊘ Warning

- ◆ Do not confuse $\geq$ with $>$.
- ◆ With mark ≥ 50 , a mark of exactly 50 passes.
- ◆ With mark > 50 , a mark of exactly 50 would fail.



Comparison

Pass

- ✓ Mark is 50 or more.
- ✓ checkPassFail() returns Pass.
- ✓ The program prints "You have passed."

Fail

- ✓ Mark is below 50.
- ✓ checkPassFail() returns Fail.
- ✓ The program prints "You have failed."

✦✦ Summary

- ✓ The program checks whether a mark is at least 50.
- ✓ An enum is used to represent Pass and Fail.
- ✓ Functions divide the program into reading, checking, and printing tasks.

Thank you 😊